NNN-T-001300 (DSA-DM)
October 31, 1969

INTERIM FEDERAL SPECIFICATION

TUBE, BLOOD COLLECTING, Vacuum, Sterile

This Interim Federal Specification was developed by the Defense Personnel Support Center, 2800 S. 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal Agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers sterile vacuum blood collecting pilot tubes.

1.2 Classification. Sterile, vacuum blood collecting pilot tubes shall be of the following types, sizes, and grades.

Type I - With additive.
Type II - Without additive.

Size 1 - 5.5 milliliters (ml.) nominal aspiration.
Size 2- 7 milliliters (ml) nominal aspiration.

Grade A - AD solution as additive.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

DD-G-541	Glass (Laboratory)
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

FSC 6630

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STANDARDS

Federal

Test Method Std. No. 601 Rubber, Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this Specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston; New York; Atlanta; Chicago; Kansas City, Mo.; Fort Worth; Denver; San Francisco; Auburn, Washington; and Washington, D.C.

(Federal Government Activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

SPECIFICATIONS

Military

MIL-L-10547 Liner, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of Military specifications and standards required by suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States (U.S.P.).

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pennsylvania 18042.)

REAGENT CHEMICALS AND STANDARDS WITH METHODS OF TESTING AND ASSAYING by Joseph Rosin

(Applications for copies should be addressed to D. Van Nostrand Company, Inc., Princeton, New Jersey 08540.)

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3. REQUIREMENTS

3.1 Material

3.1.1 Glass. Glass shall be in accordance with DD-G-541, type II.

3.1.2 Rubber. The rubber formulation shall contain at least seventy percent (70%) by volume of isobutylene - isoprene polymer and at least five percent (5%) by weight of zinc oxide. The rubber shall be non-blooming, nonbleeding and self sealing.

3.2 Construction. The blood collecting tubes shall consist of a tube and stopper. The stopper shall fit snugly in the open end of tube to form an airtight seal.

3.2.1 Tube. Tube shall be transparent and shall be fabricated of glass. Wall thickness shall be uniform. Bottom shall be round and open end shall be straight without a bead or flare. Tube shall have the following dimensions:

Length, 100 $\pm$ 2 millimeters (mm); outside diameter, 13 $\pm$ 1 mm; and wall thickness 1 + 0.1 - 0.2 mm.

3.2.2 Stopper. The stopper shall be fabricated of rubber and shall be a diaphragm, molded in one piece. Top of diaphragm shall be flat with a recessed center and bottom shall be a hollow plug. Color shall be yellow cream. Coloring material shall be inorganic and nonbleeding. Shore durometer reading of stopper shall be 50 $\pm$ 5 when tested as specified in 4.4.1.

3.3 Additives. Grade A tube shall contain 0.17 ml plus 20 percent minus 10 percent AD solution. AD solution shall contain the following ingredients in the specified proportions:

USP Citric Acid	$C_6 H_8 O_7 H_2O$	1.52 grams (gms)
USP Dextrose	$C_6 H_{12} O_6 H_2O$	5.0 gms
USP Purified Water (qs)		100.0 mls.

3.4 Sterilization assistant. Type II tube may contain 0.05 ml of isotonic sodium chloride solution to produce steam for sterilization. Sodium chloride shall be Reagent grade meeting the standards of Rosin's Reagent Chemicals and Standards and water shall be distilled.

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3.5 Performance.

3.5.1 Vacuum. When tested as specified in 4.4.2 minimum volumes of water drawn into tubes shall be as follows:

Size 1	5.0 ml.
Size 2	6.0 ml.

3.5.2 Resistance of tubes to centrifugal force. Resistance of tubes to centrifugal force shall be such that no breakage results when the tubes are tested as specified in 4.4.3.

3.6 Date of Shipment. Date of shipment shall not be more than 60 days after the date of manufacture.

3.7 Expiration date. The expiration date shall be not less than 24 months from the date of manufacture. At the expiration date the blood collecting tube shall draw at least 90 percent of the volume specified in 3.5.1.

3.8 Data sheet. If Isotonic sodium chloride is used for sterilization, each package shall contain the following statement, printed on a white sheet of paper: "May contain droplets of isotonic salt solution which will not interfere with serological tests. Not to be used for blood chemical anaylsis."

3.9 Sterility. When tested as specified in 4.4.4 interior of sealed tubes shall comply with sterility test requirements of the current edition of the U.S.P.

3.9.1 Certificate of sterility. A certificate of sterility listing all sterility lot or control number shall state that samples from each sterility lot or control number have been tested for sterility and have been found to be sterile. The certificate shall reference the method used in determining sterility for destination inspection only, one copy of certificate of sterility shall be submitted by the contractor directly to the Defense Personnel Support Center, Directorate of Medical Materiel, Attn: Chief, Quality Control Branch, Code ATQ, 2800 S. 20th Street, Philadelphia, Pa. 19101.

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3.10 Workmanship. Workmanship shall be first class throughout. Tubes shall be free of defects listed in Table III and other defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties.)

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the glass and rubber which will contain evidence of the identification of materials required in the tube may be furnished in lieu of actual tests.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105. The inspection levels and acceptability quality levels shall be as indicated in Table I. Unit of product for examination purposes shall be the tube.

Table I. Sampling for Examination

Type defect	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	0.65
Minor defects	II	2.5
For dimensional examination	S-1	2.5

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4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II. Unit of product for test purposes shall be the tube.

Table II. Sampling for Tests

Characteristic	Unit of product	Requirement	Test Procedure	Inspection level	AQL
Hardness	Stopper	3.2.2	4.4.1	S-2	*
Vacuum	Tube	3.5.1	4.4.2	S-2	1.5
Resistance to Centrifugal force	Tube	3.5.2	4.4.3	S-2	1.5
Sterility	Tube	3.9	4.4.4	**	*

* Acceptance number shall be zero.
** Sampling shall be as specified in the USP.

4.3 Examination. The tubes shall be examined for defects including but not limited to those listed in Table III.

Table III. Classification of Defects

Categories	Defects
Critical	None defined
Major	
101	Tube not free of cracks and chips.
102	Stopper not secured in tube.
103	Tube not free of dirt or foreign matter.
104	Tube does not contain additive (except type II).
Minor	
201	Tube not free of sharp edges.
202	Tube not free of bubbles.
203	Stopper not free of bloom.

4.3.1 Dimensional examination. The tubes shall be examined for defects in dimension. Any dimension not within tolerance specified herein shall be classified as a defect.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Durometer hardness. Stopper shall be tested in accordance with Method 3021 of Federal Test Method Standard No. 601.

4.4.2 Vacuum. A seamless steel tube Brown and Sharp gage No. 20, 2-1/2 ± 1/16 inches long with a point at each end and a wall thickness from 0.0055 to 0.0060 inch or FSN 6515-864-6788 shall be used for testing. One end of the testing tube or the venipuncture end of the needle shall be placed in distilled water and the other end of the testing tube or needle shall pierce the diaphragm.

4.4.3 Resistance to centrifugal force. Tubes shall be filled to capacity with distilled water and suspended in a centrifuge which has an angle head of 50° from the vertical and a radius of 1-3/4 inches from the center of rotation to the center of the mouth of the tube shield. The centrifuge shall be accelerated gradually to a speed of 3000 revolutions per minute (r.p.m.). Centrifuge shall run at 3000 r.p.m. for 30 minutes. The current shall then be shut off and the centrifuge be allowed to stop without the use of a brake.

4.4.4 Sterility. The sterility test shall be performed as described under Sterility Tests for Transfusion and Infusion Assemblies in the current edition of the U.S.P.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One package containing 100 tubes, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table IV. Packaging Quantities

Type	Unit package	Intermediate package	Exterior container
I and II	1 unit	Not Required	10 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a corrugated fiberboard box of appropriate size and design fabricated of 200 pound test fiberboard. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with paragraph 30.4 of PPP-B-636.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit packages. Each unit package shall be marked in accordance with MIL-STD-129. Sterility lot control number, contract or purchase order number, name of contractor and the following legend shall be shown: "STORE IN A COOL PLACE (50° F-80° F)". Type II Shelf-Life markings as specified in 5.2.2.1 (b) of MIL-STD-129 shall be shown. The "INSPECTION/TEST DATE" and "TEST NO." marking shall include a blank space for inclusion of data at a later date. In addition, the method of sterilization (G-GAS, R-RADIATION, S-STEAM, D-DRY HEAT STERILIZATION) shall be shown following the level of packaging and date of packaging. Example: "A-9/69-G". Note: Final sterilization shall be on or after indicated packaging date. A printed sheet shall be included in each unit package containing the following statement: "MAY CONTAIN DROPLETS OF ISOTONIC SALT SOLUTION WHICH WILL NOT INTERFERE WITH SEROLOGICAL TESTS. NOT TO BE USED FOR BLOOD CHEMICAL ANALYSES".

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Sterility lot control number and the following legend shall be shown: 'STORE IN A COOL PLACE (50° F - 80° F)". Type II Shelf-Life markings as specified in 5.2.2.1 (b) of MIL-STD-129 shall be shown. The "INSPECTION/TEST DATE" and "TEST NO." marking shall include a blank space for inclusion of data at a later date. In addition, the method of sterilization (G-GAS, R-RADIATION, S-STEAM, D-DRY HEAT STERILIZATION) shall be shown following the level of packaging and date of package. Example: "A-9/69-G". Note: Final sterilization shall be on or after indicated packaging date.

NOTE: The sterility lot control number shall be traceable to each sterilized charge.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, name, and date of this specification.
b. Type, size and grade (if applicable).
c. Selection of applicable packaging and packing (See 5.1 and 5.2).
d. Federal Stock Number.

6.2 This specification covers the following items listed in the Federal Supply Catalog:

Federal Stock No.	Item Identification	Type	Size	Grade
6630-145-1525	Tube, Blood Collecting, Vacuum, 7 ml without Anticoagulant, 100s	II	2	None
6630-145-1526	Tube, Blood Collecting, Vacuum, 5.5 ml, with Citric Acid and Dextrose Solution, without Anti-coagulant, 100s	I	1	A

6.3 The items listed in 6.2 are to be used as pilot tubes with the following items listed in Federal Supply Catalog:

Federal Stock No.	Item Identification
6515-926-4766	Blood Collecting-Dispensing Bag and Donor Set
6515-926-4767	Blood Collecting-Dispensing Bag and Donor Set, Double Bag

6.4 Specified rubber is the only known material which is compatible with sampling and analysis and provides sufficient seal to hold the vacuum over an extended period of time. Other rubber will be considered if detailed reports on the effects on sampling and analysis and the durability of the seal are submitted.

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6.5 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Federal Government requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities
Army - MD
Navy - MS
Air Force- 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements

SPECIFICATION

NNN-T-001300(DSA-DM) FSN

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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